Problem Statement

The problem our prototype addresses is indoor wayfinding inside Patrick F. Taylor (PFT) Hall. PFT is a large, multi-floor academic building with many classrooms, laboratories, faculty offices and shared spaces distributed across several sections of the building. Because of its size and complexity, students and visitors can easily become confused while trying to locate a destination, especially when they are unfamiliar with the building layout. Many hallways look similar, some destinations are placed far from major entrances, and moving between floors adds another layer of difficulty. As a result, users often spend extra time wandering, checking room numbers repeatedly, or asking others for directions. Our earlier design work showed that this problem is not only about finding a room number on a map. Users also need help understanding where they are in relation to the destination, which floor they should be on, and what path they should follow through the building. In other words, they need both destination search support and navigation support. A static building map alone is often not enough, because it does not actively guide users through the route or help them recover when they are uncertain. This is especially true for first-time visitors, new students, or anyone visiting a department or lab that they do not regularly access. Therefore, the goal of our final prototype is to provide a practical indoor navigation system that helps users search for a destination and navigate to it through a simple, floor-aware interface. The prototype focuses on making indoor navigation faster, clearer, and less stressful by combining destination lookup, destination confirmation, route visualization and route-state feedback into a single system. Rather than overwhelming the user with too much information at once, the design aims to provide clear guidance at the moment it is needed. In this way, the prototype responds directly to the central problem of indoor confusion in PFT and offers a more usable alternative to relying only on posted signs or static floor plans.

Four Primary Features Implemented: To address this problem, our final prototype implements four main features, organized around the two major tasks identified in our earlier deliverables: **searching for a destination** and **navigating to the destination**.

Task 1: Search for a Destination

1. Searchable destination lookup

Figure 1: The search interface supports keyword-based suggestions; entering “pa” returns “Panera” as a suggested destination

The first feature is a searchable destination lookup system. The prototype includes a search-based interface that allows users to enter a room number, office number or other identifying information in order to find a destination (see figure 1). This feature is important because many users know what they are looking for, but not where it is located in the building. For example, a student may know that they need to go to panera or room number 1375, but they may have no idea which part of PFT that room is located in. The search feature reduces this difficulty by letting the user to begin with familiar information rather than forcing them to manually inspect the entire floor plan.

This feature improves efficiency because it shortens the first step of navigation. Instead of scanning multiple hallways or comparing room labels visually the user can enter the target directly and immediately begin the process of locating it. This is especially useful in time-sensitive situations such as when a student is trying to reach a classroom before class starts or when a visitor is attempting to find an office for a meeting. By allowing quick destination entry the prototype lowers the mental effort required to begin navigating.

2. Destination selection and confirmation

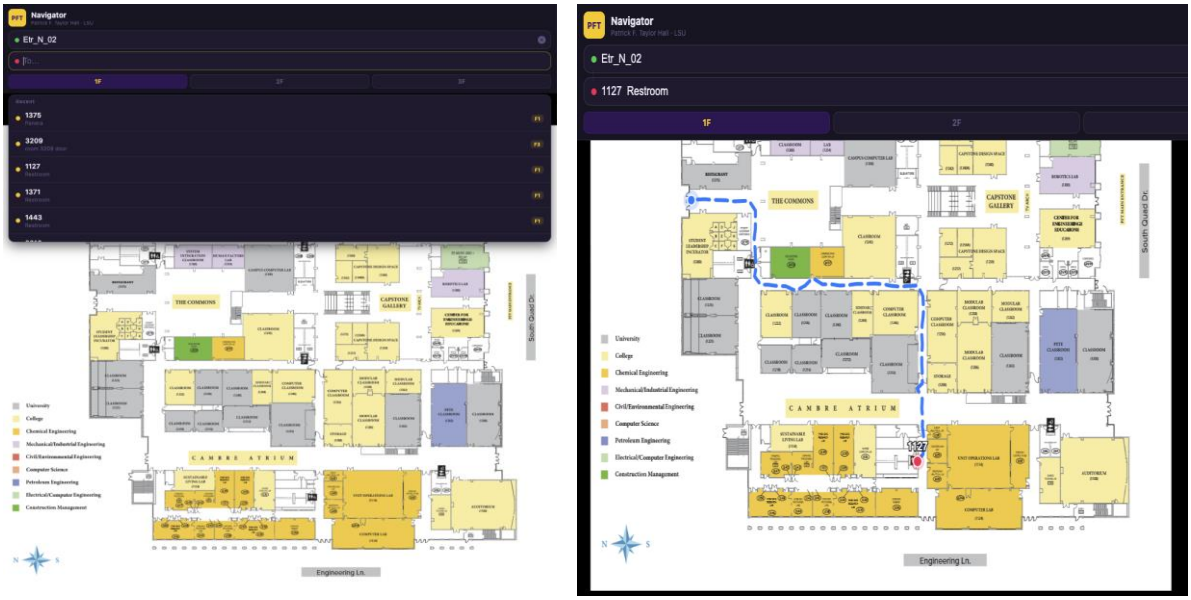


Figure 2: Destination lookup interface showing previously searched rooms (left) and destination selection for Restroom 1127 with the resulting navigation path (right).

The second feature is destination selection and confirmation (see Figure 2). After a user searches for a location the system helps them verify that they selected the correct destination before beginning navigation. In the current interface, the destination card displays information such as the room, label, and floor and it also allows the user to clear the current selection if necessary. This feature may seem simple, but it is very important in a building where users can easily confuse similar room numbers, departments or locations. This confirmation step reduces user error. A user should not begin following a route if they are unsure whether they selected the correct place. By clearly showing destination details the prototype gives the user a moment to review the information and make sure it matches their intention. This creates stronger sense of control and lowers the chance of being guided to the wrong location. In practice, this feature supports confidence as much as accuracy: users are more likely to trust the system when they can clearly see what destination has been selected.

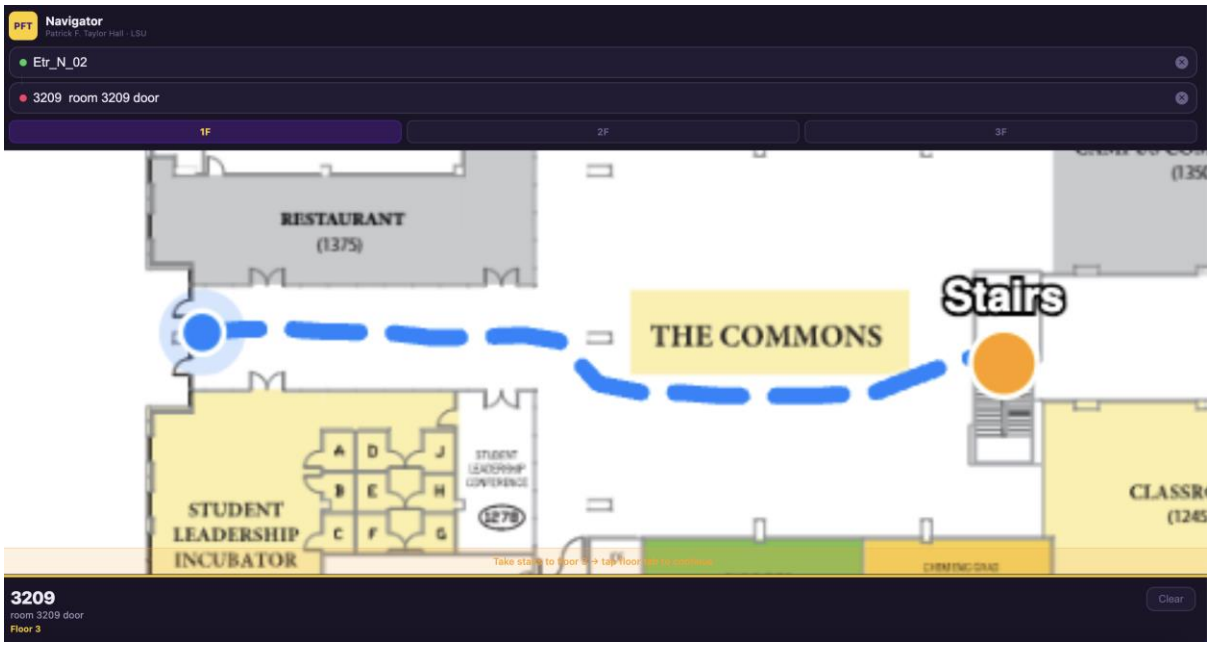
Task 2: Navigate to the Destination

3. Interactive indoor map with route visualization: The third feature is the interactive indoor map with route visualization. Once a destination is selected the system presents a floor-based map and highlights the route that the user should follow. This feature is central to the purpose of the prototype because it transforms the interface from a simple search tool into an actual navigation aid. A highlighted route helps users understand not only where their destination is but also how to move through the building's hallways and intersections to reach it. This feature is especially important in PFT because spatial confusion is one of the main causes of navigation difficulty. Users may understand that room exists somewhere on the given floor but still struggle to understand how to physically go there. Route visualization addresses this gap by showing visible path rather than leaving the user to interpret the map on their own. The prototype also supports multiple floors with separate floor tabs such as 1F, 2F and 3F. This makes the system more realistic for a building like PFT where users often need to think not only about horizontal movement through hallways but also about vertical movement between floors. As a result the map becomes a dynamic guide rather than a passive reference image.

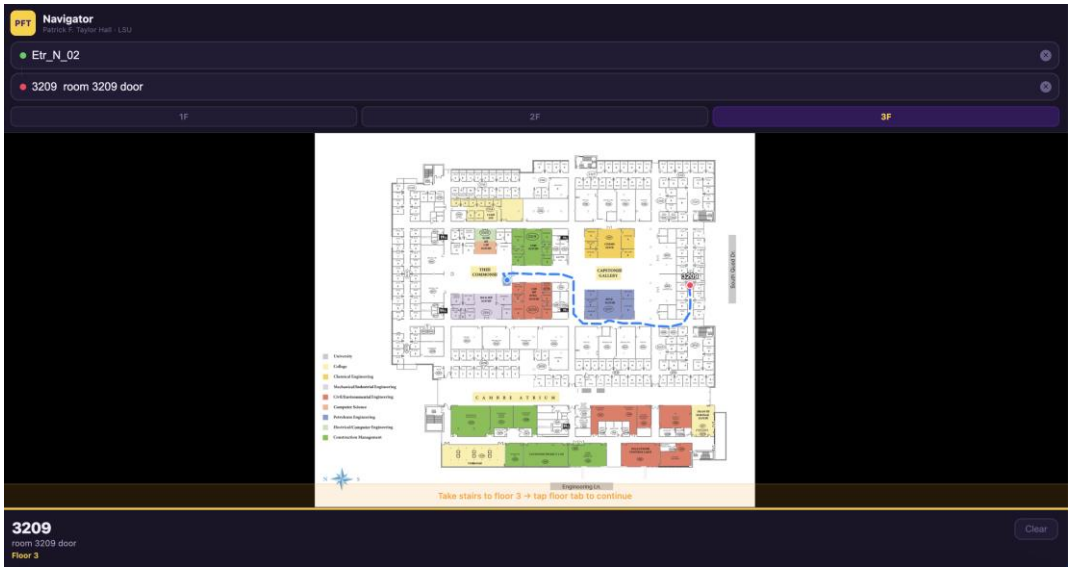
4. Cross-floor and route-state guidance: This feature supports navigation across multiple floors while providing clear system feedback. When a route involves floor changes, the system guides users in steps: first to the nearest staircase or elevator, then from the corresponding point on the destination floor to the final room. It also includes route-state messages such as loading, "no route found," and destination status.

This improves clarity and reduces confusion by clearly indicating floor transitions and system status.

Step 1: From Floor 1, navigate to the nearest stairs/elevator.



Step 2: From Floor 3 corresponding stair, continue to Room 3209.



Challenges Faced During Design and Development

One of the biggest challenges we faced while developing the PFT indoor navigation system was that the entire prototype had to work as a frontend-only system. We did not use a backend server, and we also could not rely on real-time indoor positioning because GPS is often weak or unavailable inside buildings. As a result, the system could not determine a user's exact live location the way outdoor navigation systems typically do. This meant we had to build the navigation logic using raster floor maps, normalized coordinates, and static JSON-based room and graph data instead of relying on a real-time positioning service. Another major challenge was data quality. The routing algorithm itself was comparatively straightforward, but the real difficulty was making sure that the node coordinates, room mappings and edge connections were accurate. Even a small error in annotation could produce incorrect routes, broken navigation, or missing destinations. Because of this, preparing the data became one of the most time-consuming parts of the project. It also required building a dedicated annotation tool from scratch so that rooms, doors, and connections could be marked consistently across all three floors. A related technical challenge was that automatic edge connection was not always reliable. If we relied only on distance thresholds, some nodes could be connected through walls even when they were not actually reachable in the physical building. This created invalid paths and reduced routing accuracy. In addition after importing graph data new nodes could sometimes create ID collisions which would break routing logic and make the graph inconsistent. Hence the challenge was not only creating the map but also making sure the underlying navigation structure remained valid as the system evolved. Another challenge was iterative nature of development. Both the annotation tool and main navigation application went through multiple revisions because many issues only became visible during testing. Some design and technical decisions that looked reasonable at first had to be revised later when they caused problems in routing, node management or map usability. This made development less linear and also required repeated debugging, testing and redesign. Finally, we also faced the challenge of handling uncertainty and failure states in the interface. In real use the system does not always move directly from search to success. Sometimes a route may not be found, the system may still be loading, or the user may need to clear and restart the destination. If the system does not communicate these states clearly users may assume that it is broken or they can become confused about what to do next.

Overcoming Challenges:

We overcame the frontend-only and no-GPS limitation by designing the system around static floor maps, normalized coordinates, and structured JSON files instead of real-time positioning. Rather than trying to detect live indoor location, we focused on building a floor-aware pathfinding model in which rooms, doors, hallways, stairs, and transition points were all represented inside the graph structure. This allowed the prototype to provide meaningful indoor navigation even without relying on GPS or backend services. To address the data-quality problem, we built a dedicated annotation tool and used it to manually define nodes, destinations, and route connections. This made it possible to control how the floor maps were translated into usable navigation data. We also kept the floor maps aligned and consistent so that the routing structure remained stable across all three floors. Although manual annotation took significant effort it allowed us to produce a working prototype with accurate enough routing for demonstration and evaluation. We addressed the auto-connection problem by using different threshold rules for different kinds of connections, such as corridor-to-corridor connections versus door-to-corridor connections, instead of relying on a single distance rule for everything. This reduced invalid edges especially those that would otherwise pass through walls. To prevent node ID collisions after import we added a way to sync the counters with the highest existing node IDs before creating any new ones. This helped us to keep the graph structure consistent. Also it prevented routing problems after data was imported. We handled the iterative development challenge by treating the project as a process of continuous refinement rather than expecting the first version to be final. Both the annotation tool and the main app were updated several times based on testing and debugging. This process helped us to spot issues in the data, routing and interface design and improve them gradually until the system became more stable and usable. To handle uncertainty and failure cases we added route-state feedback directly into the interface. Features such as the loading panel, no-route message, clear destination button and cross-floor banner were included. As a result users could understand what the system was doing at each stage. These features improved clarity and reduced the chance that users would feel lost or assume the prototype was not functioning properly.

● If and how did your features change from the low-fi prototype described in milestone 2?
Yes, the core tasks remained the same, but several features were refined from the low-fidelity prototype to the final prototype. In Milestone 2, the low-fi design emphasized a search bar, location history, an indoor map with route visualization and a separate step-by-step navigation view. In the final prototype, the search feature was streamlined into a direct search input with a results panel and destination card, which made destination selection and confirmation more central than search history. Similarly, the map-based route visualization was retained, but the separate step-by-step instruction concept appears to have been simplified into floor-aware navigation support through floor tabs, a cross-floor banner, loading and no-route states, and destination feedback. In addition, the final prototype introduced technical backend work, including aligned floor maps, structured room and graph data, and an annotation workflow, which made the system more functional than the earlier low-fi concept.
Methods we use to evaluate our prototype: We used two ways to test: a heuristic evaluation and semi-structured user testing sessions. Two HCI graduate students who did not work on the prototype design looked at the interface on their own using Nielsen's 10 heuristics for the heuristic evaluation. They looked at how well the prototype followed each heuristic and gave it a severity rating from 0 (not a usability problem) to 4 (catastrophic problem). This method helped us find usability problems in a systematic way without having to hire end users. During the user testing sessions, we watched five people do four important things with the prototype: (1) find a destination, (2) choose and confirm a

destination, (3) follow a route on the map, and (4) move between floors. We asked follow-up questions after each task. We looked at how long it took to finish the task, how many mistakes were made, and how hard it seemed.

During the user testing sessions, we watched five people do the same four tasks and asked them follow-up questions. We looked at how long it took to finish a task, how many mistakes were made, and how hard it seemed.

Why did we choose this method: We picked these two methods for different reasons. First, heuristic evaluation is a quick way to find usability problems that are already known. Experts can look at the interface without having to get people to do it, which saves time and money. Second, user testing shows how real people act in real situations. Seeing how real PFT users used the prototype showed problems that experts might not have noticed, like the exact times when participants missed cues for floor transitions. Using both methods together gave us a full picture. The heuristic evaluation found design mistakes that went against established rules. The user testing showed us how people act when they are stressed, in a hurry, or confused. This mixed method is common in HCI research because expert inspections and user studies find different types of problems.

Participants: We observed five people who spend a lot of time in Patrick F. Taylor Hall. Everyone was asked to join through word of mouth and requests made in person around the building. People could choose to take part, and they weren't paid.

We picked a group of students who all had different levels of technology knowledge and PFT familiarity. P1 and P4 are less experienced PFT users who still get lost. P2 and P5 are power users who know how to get around but still notice problems. P3 is a teaching assistant too, who might use the system to help others. This variety helps us find out if the prototype works for everyone or just for some people.

Procedure: The sessions lasted about 25 to 30 minutes and were held at the entrance of PFT. We temporarily hosted our prototype and ran it on a mobile phone so that participants could use the system in the same building where they usually navigate. We told everyone at the start of each session what the study was about and asked for their verbal permission. After that, we asked them to do four things with the mobile prototype. We didn't tell them what to do step by step. Instead, we gave them real-life situations and watched how they acted in them. The four things to do were:

Task 1: "You need to find any room of your choice on the first floor (e.g., Room1236)." Find it with the system.

Task 2: "You chose room. Now check that it is right and begin to move.

Task 3: "Use the map to find your way to the room." As you go, say out loud what you're thinking.

Task 4: "Now go to a room on the third floor. Listen to what the system says to do when you want to change floors.

We asked two quick questions after each task: "On a scale of 1 to 5, how hard was that?" and "Was there anything hard to understand?" After all, four tasks were done, we asked three questions to wrap things up: what was the most confusing part, if they would really use the system, and what one thing they would change. We used a stopwatch to see how long it took to finish a task. One of the researchers used the stopwatch. A second researcher wrote down where participants clicked, what they said out loud, and what mistakes they made. With permission, we recorded each session on audio.

Analysis:

We employed two analytical methods. We averaged the task completion times and difficulty ratings for the five participants to get the quantitative data. We used thematic analysis for the qualitative data, which included comments from participants, errors we saw, and answers given after the session. One researcher read all of the session notes and audio transcripts and then found themes that came up again and again. Another researcher looked over the themes, and any disagreements were talked about until they were settled. Everyone found Task 1 and Task 2 to be quick and easy, taking less than 20 seconds

each. It took an average of 12 minutes and 30 seconds for people to do Task 3, which required them to physically walk the route inside PFT while following the prototype. It took the longest time (14 minutes and 20 seconds) to complete Task 4, which involved finding a room on a different floor. It was also rated the hardest. P3 was the slowest of all the walkers, taking almost 16 minutes to walk across the floor. P2 was the fastest, finishing Task 4 in just over 12 minutes.

Quantitative Results: Table 2 presents average task completion times and difficulty ratings across the five participants.

Task	**Average Time (min:sec)**	**Average Difficulty (1-5)**
Search for destination	18 seconds	2.0
Confirm destination	11 seconds	1.6
Follow route on one floor	12 minutes 30 seconds	3.0
Cross-floor navigation	14 minutes 20 seconds	3.6

The search and confirmation tasks were done quickly, each taking less than 20 seconds and getting low difficulty ratings. Route-following tasks took a lot longer. For example, cross-floor navigation took an average of 14 minutes and 20 seconds and got the greatest difficulty rating (3.6). P3 was the slowest at walking tasks, taking 15 minutes 50 seconds to cross the floor, and P2 was the fastest, taking 12 minutes 05 seconds.

Qualitative Results: Thematic analysis of participant interviews identified four themes that delineate users' experiences with the prototype.

Theme 1: How well the destination lookup works: Participants were able to find and confirm their destinations with little trouble. All five people finished the search and confirmation tasks without making any mistakes. P2 said, “The search bar is right there. I didn't have to look for it.” P5 mentioned, “The destination card shows me what I chose. This helps me double-check because I have definitely typed the wrong room number before.”

Theme 2: No feedback on where you are: While following routes, participants always had trouble figuring out where they were on the map. P4 said, “Once I know where I am, it's easy to follow the route line." P3 similarly noted, "I see the line, but I still have to stop and look at the map at each corner. It doesn't say, 'turn left in 20 feet’ as Google Maps does.

Theme 3: Floor transition cues are hard to see: Without any specific prompting, four out of five participants did not notice the cross-floor banner or floor tabs. “I didn't see the 3F tab at all,” P1 said. I kept looking at the 2F map and wondering why the route was gone. P3 said, “I only noticed the floor tabs because you told me to pay attention to changes in the floor.” P2 said, “I might not notice if I were using this on its own.”

Theme 4: Need for Passive Guidance Methods: Participants said they would rather have auditory or turn-by-turn directions than a visual map. P4 said, “The map is fine, but I don't want to look at my phone while I'm walking. I want it to say Turn left now.” P3 elaborated, "In a building, this is confusing; I need audio. When I look at a map, I have to stop walking, and when I'm already late, stopping makes me more stressed.

Overall, four people said that the most confusing part of the prototype was the transitions between floors when they were asked. One participant said that there wasn't a marker for the current location. Four people said they would use the system if it were available, but P3 said, “Yes, but only if it had voice directions. If I didn't have voice, I would just ask someone.” Three people asked for a “you are here”

marker, two asked for voice guidance, and one asked for bigger floor tabs when asked to name one thing that could be better.

Summary of Heuristic Evaluation: Two of the graduate students independently used Nielsen's 10 heuristics to test the prototype. Table 3 shows what they found.

Heuristic	**Severity (0–4)**	**Finding**
System status is visible	1	Loading state is present but not obvious
Match between system and real world	0	Room numbers and floor labels match PFT
User control and freedom	1	Clear destination button works; no undo function
Consistency and standards	1	Floor tabs consistent across all floors
Error prevention	2	No validation for misspelled or invalid room numbers
Recognition rather than recall	1	Destination card reduces memory load
Flexibility and efficiency of use	2	No shortcuts for frequent users
Minimalist design	0	Interface clean and uncluttered
Help users recognize and recover from errors	2	"No route found" message appears; no recovery guidance
Help and documentation	3	No help section or onboarding tutorial

The evaluators found that the most serious problem was missing documentation (Heuristic 10). “A first-time user who has never seen this app before will have to figure everything out by clicking around,” one evaluator said. This could be fixed with just one short tutorial page. The second evaluator said, “The missing 'you are here' marker is not technically a heuristic violation, but it is the single most important usability issue in practice.”
The evaluation showed that the prototype worked well for searching for and confirming destinations. However, three major usability problems came up: (1) the lack of self-location feedback made following the route mentally taxing, (2) floor transition cues weren't clear enough, and (3) participants wanted passive guidance methods like voice directions. These results give clear guidance for making iterative design improvements.

3. Participation:

Sl	Participant Name	Email	Contribution Areas
1	Tania Khatun	tkhatu1@lsu.edu	Brainstorming, development, wrote final prototype section
2	FNU Nurjahan	nurja1@lsu.edu	Brainstorming, development, wrote final prototype section
3	Nushrat Jahan Ria	nria1@lsu.edu	Brainstorming, development, wrote evaluation section
4	Sidhi Kumari Yadav	SidhiKumari.Yadav@lsu.edu	Brainstorming, development, wrote evaluation section
5	Ming Li	Ming.Li@lsu.edu	Brainstorming, development, wrote final prototype section